

MAT21-2

Probability & Statistics I

Albert School — 34 hours

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Preface

In **Foundations** (MAT11-1) you built the language of calculus and rigorous reasoning: set notation and logical connectives, sums written with $\sum$, functions and their limits, derivatives and the way a first-order condition $f'(x) = 0$ locates a maximum, the exponential and logarithm and their functional equations, geometric series $\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$ for $|r| < 1$, and the small-argument approximations $e^x \approx 1 + x$ and $\ln(1 + x) \approx x$. In **Linear Algebra I** (MAT11-3) you learned to read and write short proofs — direct verification, counterexamples, if-then statements — and to think of structured objects as living in a space. This book leans on both constantly. We will **differentiate a log-likelihood** to find an estimator, **sum a geometric series** to find an

expectation, and **verify axioms** the way you once verified the subspace test. If a manipulation here feels unfamiliar, the gap is almost always in MAT11-1, not in probability.

Probability is the part of mathematics that reasons about **uncertainty** without pretending it away. A fair coin will land heads or tails — we cannot say which, yet we can say a great deal: how often heads occurs in the long run, how surprised we should be by ten heads in a row, how a positive medical test should change our belief about being ill. The central skill this course builds is **modelling**: turning a sentence of ordinary language — “a factory’s machine produces a defective part 2% of the time; in a box of 50, what is the chance of at least one defect?” — into a precise mathematical object you can compute with. That skill, **reasoning over recognition**, matters more than any single formula. A student who memorises the binomial formula but cannot tell whether a problem is binomial has learned nothing durable; a student who can build the model can re-derive the formula whenever needed.

Our method, throughout: every idea arrives as a **concrete example first** and the **general statement second**. We flag the mistakes students reliably make — they are predictable, and naming them is half of avoiding them. And we close the course where modern data analysis begins: with **estimation** (reading a parameter off data) and **Bayesian updating** (revising a belief as evidence arrives).

1 Counting

Before we can assign probabilities we must **count**. When every outcome of an experiment is equally likely — a fair die, a well-shuffled deck — the probability of an event is simply the number of outcomes in it divided by the total number of outcomes. So “how probable” reduces to “how many”, and counting becomes the first tool of probability.

1.1 The multiplication principle

Suppose a café offers 3 sizes of coffee and 4 syrups. How many distinct drinks can you order? For each of the 3 sizes you may pick any of the 4 syrups, giving $3 \times 4 = 12$ drinks. Nothing about coffee is special here.

Theorem 1 (Multiplication principle) If a procedure consists of k successive choices, where the first can be made in n_1 ways, the second in n_2 ways (whatever the first was), and so on, then the whole procedure can be carried out in

$$n_1 \times n_2 \times \cdots \times n_k$$

ways.

The phrase “whatever the first was” matters: the **number** of options at each stage must not depend on the earlier choices, even though the options themselves may.

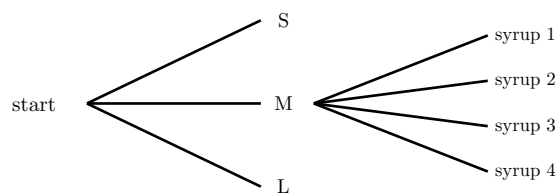


Figure 1: The multiplication principle as a tree. From the “M” size, four syrups branch off; the same four hang from “S” and “L”. Counting the final leaves gives $3 \times 4 = 12$ — multiply the options at each stage.

1.2 Permutations and arrangements

Worked example — seating three friends. In how many orders can Ana, Bo and Cy sit on a bench? The first seat has 3 candidates; once filled, the second has 2; the last has 1. By the multiplication principle, $3 \times 2 \times 1 = 6$ orders.

An ordering of all the objects in a set is a **permutation**. The count $3 \times 2 \times 1$ is written $3!$ (“3 factorial”).

Definition 2 For an integer $n \geq 0$, the **factorial** is $n! = n \times (n-1) \times \cdots \times 2 \times 1$, with the convention $0! = 1$. The number of **permutations** (orderings) of n distinct objects is $n!$.

Now suppose we do not arrange **all** the objects but only some of them, **with order mattering**. From 10 runners, how many distinct gold–silver–bronze podiums are possible? Gold: 10 choices; silver: 9; bronze: 8; total $10 \times 9 \times 8 = 720$.

Definition 3 (arrangement / \$k\$-permutation) The number of ordered selections of k distinct objects from n (an **arrangement**, or k -permutation) is

$$A_n^k = n \times (n-1) \times \cdots \times (n-k+1) = \frac{n!}{(n-k)!}.$$

Common pitfall Order is the whole question. “How many podiums?” counts arrangements (gold $\neq$ silver). “How many three-runner committees?” — order irrelevant — counts **combinations**, the next idea. Deciding **does order matter here?** before reaching for a formula is the single most important counting habit.

1.3 Combinations and the binomial coefficient

From 10 runners, how many **unordered** groups of 3 can we pick (say, to share a prize equally)? The 720 ordered podiums overcount each group of 3, because the same three runners can be ordered in $3! = 6$ ways. So the number of groups is $\frac{720}{6} = 120$.

Definition 4 (combination / binomial coefficient) The number of ways to choose k objects from n **without regard to order** is the **binomial coefficient**

$$\binom{n}{k} = \frac{A_n^k}{k!} = \frac{n!}{k!(n-k)!}, \quad 0 \leq k \leq n.$$

We read $\binom{n}{k}$ as “ n choose k ”.

Two facts are immediate from the formula and worth knowing by reflex:

$$\binom{n}{k} = \binom{n}{n-k} \quad (\text{choosing what to keep} = \text{choosing what to drop})$$

and $\binom{n}{0} = \binom{n}{n} = 1$. A third, **Pascal’s rule**, builds each coefficient from two above it.

Proposition 5 (Pascal’s rule) For $1 \leq k \leq n-1$,

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Proof. Fix one particular object, say object $\star$. A k -subset of the n objects either contains $\star$ or not. Those containing $\star$ are completed by choosing $k-1$ of the remaining $n-1$ objects: $\binom{n-1}{k-1}$ of them. Those omitting $\star$ choose all k from the other $n-1$: $\binom{n-1}{k}$. The two cases are disjoint and exhaustive, so their counts add. $\square$

This proof is worth pausing on: we counted the **same** set in two ways and equated the answers. That “story” style of proof — **combinatorial proof** — is often shorter and clearer than algebra, and it explains **why** a identity holds.

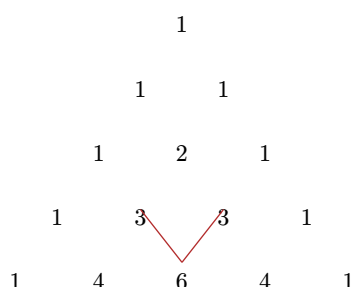


Figure 2: Pascal’s triangle: row n lists $\binom{n}{0}, \dots, \binom{n}{n}$. Each entry is the sum of the two above it (Pascal’s rule). The red lines show $\binom{3}{1} + \binom{3}{2} = 3 + 3 = 6 = \binom{4}{2}$.

The name “binomial coefficient” comes from the **binomial theorem**, which you can now read combinatorially: expanding $(a+b)^n = (a+b)(a+b)\cdots(a+b)$ means picking a or b from each of the n factors; the term $a^k b^{n-k}$ arises exactly when we pick a from k of the factors, which happens in $\binom{n}{k}$ ways. Hence

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}.$$

Setting $a = b = 1$ gives $\sum_{k=0}^n \binom{n}{k} = 2^n$: the total number of subsets of an n -set. We will reuse this identity to check that the binomial distribution’s probabilities sum to 1.

1.4 Multinomial coefficients: partitioning into groups

Combinations split n objects into **two** groups (the k chosen and the $n-k$ left). What if we split into **several** labelled groups at once?

Worked example — dealing a deck into hands. We deal 12 distinct cards into three labelled hands — North gets 5, East gets 4, West gets 3. How many deals? Choose North’s 5 from 12: $\binom{12}{5}$. Then East’s 4 from the remaining 7: $\binom{7}{4}$. West takes the last 3: $\binom{3}{3} = 1$. The total is

$$\binom{12}{5} \binom{7}{4} \binom{3}{3} = \frac{12!}{5!7!} \cdot \frac{7!}{4!3!} \cdot 1 = \frac{12!}{5!4!3!} = 27720.$$

The intermediate factorials telescope, leaving one clean expression.

Definition 6 (multinomial coefficient) The number of ways to partition n distinct objects into labelled groups of sizes $k_1, k_2, \dots, k_m$ (with $k_1 + \dots + k_m = n$) is

$$\binom{n}{k_1, k_2, \dots, k_m} = \frac{n!}{k_1! k_2! \dots k_m!}.$$

The binomial coefficient is the case $m = 2$: $\binom{n}{k} = \binom{n}{k, n-k}$.

A close cousin is counting **permutations with repeated, indistinguishable objects**. How many distinct words can we form from the letters of MISSISSIPPI (1 M, 4 I, 4 S, 2 P, 11 letters total)? If all letters were distinct there would be $11!$ orderings, but the 4 I's can be permuted among themselves in $4!$ ways without producing a new word, and likewise for the S's and P's. The count is

$$\frac{11!}{1! 4! 4! 2!} = 34650,$$

exactly the multinomial coefficient $\binom{11}{1, 4, 4, 2}$. The two viewpoints — “partition positions into groups by letter” and “divide out indistinguishable rearrangements” — are the same count.

Exercises

1. A password is 4 characters, each a digit (0–9). How many are possible if repetition is allowed? If not?
2. A committee of 4 is chosen from 7 women and 5 men. How many committees have exactly 2 women? At least 3 women?
3. Show $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ algebraically from the factorial formula, and compare with the combinatorial proof above.
4. How many distinct arrangements of the letters of BANANA are there?

2 Finite probability spaces

Counting tells us **how many** outcomes. Probability theory tells us how to **assign weights** to them so that “how likely” has a precise meaning. The framework is due to Kolmogorov and is astonishingly simple: three axioms generate everything.

2.1 Sample space and events

Worked example — rolling a die. Roll one fair six-sided die. The set of **possible outcomes** is $\Omega = \{1, 2, 3, 4, 5, 6\}$. The statement “the roll is even” corresponds to the **subset** $\{2, 4, 6\} \subseteq \Omega$. We will call such subsets **events**.

Definition 7 A **sample space** Ω is the set of all possible outcomes of an experiment. An **event** is a subset $A \subseteq \Omega$. The outcome ω **realises** the event A when $\omega \in A$. We call Ω the **certain event** and $\emptyset$ the **impossible event**.

Because events are sets, the logical connectives of MAT11-1 become set operations. “**A and B**” is the intersection $A \cap B$; “**A or B**” (inclusive) is the union $A \cup B$; “**not A**” is the complement $A^c = \Omega \setminus A$. The events A and B are **mutually exclusive** (disjoint) when $A \cap B = \emptyset$ — they cannot both occur.

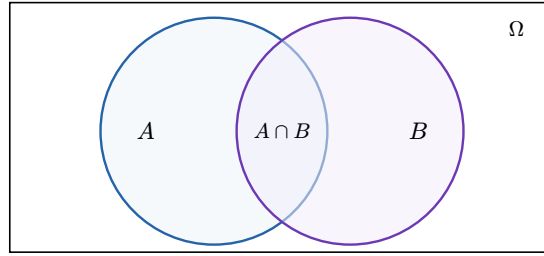


Figure 3: Events as subsets of the sample space Ω . The lens-shaped overlap is $A \cap B$ (“both occur”); the whole shaded region is $A \cup B$ (“at least one occurs”). Everything outside A is A^c .

For **finite** sample spaces we take every subset to be an event; the collection of all events is then the power set of Ω . (In MAT21-2 all sample spaces are finite or countable, so we never worry about which subsets are “allowed”. The general object, a **sigma-algebra** — a collection of events closed under complements and countable unions — is mentioned for vocabulary and developed when continuous probability arrives.)

2.2 The probability axioms

Definition 8 (probability measure) A **probability** on a finite sample space Ω is a function P assigning to each event A a number $P(A)$ such that

1. (**non-negativity**) $P(A) \geq 0$ for every event A ;
2. (**normalisation**) $P(\Omega) = 1$;
3. (**additivity**) if A and B are disjoint, $P(A \cup B) = P(A) + P(B)$.

The triple $(\Omega, \mathcal{P}(\Omega), P)$ is a **probability space**.

From these three rules every other property follows by pure logic — no new assumptions. This is exactly the proof literacy of MAT11-3 at work.

Proposition 9 For any events A, B : (a) $P(A^c) = 1 - P(A)$; quad (b) $P(\emptyset) = 0$; quad (c) if $A \subseteq B$ then $P(A) \leq P(B)$; quad (d) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Proof. (a) A and A^c are disjoint with union Ω , so by axioms 3 and 2, $P(A) + P(A^c) = P(\Omega) = 1$. (b) Take $A = \Omega$ in (a). (c) Write $B = A \cup (B \setminus A)$, a disjoint union, so $P(B) = P(A) + P(B \setminus A) \geq P(A)$ since the last term is ≥ 0 . (d) Split $A \cup B$ into the disjoint pieces A and $B \setminus A$; also $B = (A \cap B) \cup (B \setminus A)$ disjointly. Subtracting eliminates $P(B \setminus A)$ and leaves the claim. $\square$

Property (d) is the **inclusion–exclusion** principle: adding $P(A)$ and $P(B)$ double-counts the overlap $A \cap B$, so we subtract it once.

Common pitfall A frequent error is to write $P(A \cup B) = P(A) + P(B)$ in general. That is only valid when $A \cap B = \emptyset$. When the events can co-occur, you **must** subtract $P(A \cap B)$.

2.3 Equiprobability and the link to counting

Definition 10 (uniform probability) If Ω is finite and every outcome is equally likely, then for each event A ,

$$P(A) = \frac{|A|}{|\Omega|} = \frac{\text{number of favourable outcomes}}{\text{number of possible outcomes}}.$$

This is the only place counting and probability meet directly — but it is a place we visit often. Every combinatorial result of Chapter 1 is now a probability tool.

Worked example — two aces in a poker hand. Five cards are dealt from a standard 52-card deck. What is the probability that **exactly two** are aces? The sample space is all $\binom{52}{5}$ hands, each equally likely. A favourable hand chooses 2 of the 4 aces and 3 of the 48 non-aces: $\binom{4}{2}\binom{48}{3}$ of them. Hence

$$P = \frac{\binom{4}{2}\binom{48}{3}}{\binom{52}{5}} = \frac{6 \times 17296}{2598960} \approx 0.0399.$$

Notice the modelling move: we **chose** Ω to be unordered hands so that outcomes are equiprobable, then counted favourable cases with combinations.

Common pitfall Equiprobability is an **assumption about the model**, not a law of nature. The sum of two dice is **not** uniform on $\{2, \dots, 12\}$ — 7 is far likelier than 2 — because the equally likely objects are the 36 **ordered pairs**, not the 11 sums. Always identify the level at which outcomes are genuinely equally likely.

Exercises

5. From the die, let $A = \{\text{even}\}$ and $B = \{\text{at least } 4\} = \{4, 5, 6\}$. Compute $P(A)$, $P(B)$, $P(A \cap B)$, $P(A \cup B)$ and check property (d).
6. Two fair dice are rolled. Using the 36 ordered pairs as Ω , find the probability that the sum is 7; that the sum is at least 10.
7. A box has 5 red and 3 blue balls; 3 are drawn without replacement. Find the probability that all three are red, and that exactly one is blue.

3 Conditional probability and Bayes' theorem

So far every probability has been assessed before any information arrives. The heart of inference is the opposite: **given that I know B occurred, how should I re-weight A ?** This is **conditional probability**, and it leads directly to Bayes' theorem — the engine of belief revision.

3.1 Conditioning as restricting the sample space

Worked example — a die, with a hint. You roll a fair die but only learn that the result is even — the event $B = \{2, 4, 6\}$. Given this, how probable is $A = \{\text{the roll is } 6\}$? The world has shrunk to the three even faces, each still equally likely, so the chance is $1/3$, not $1/6$. Conditioning **renormalised** the probabilities onto B .

Definition 11 For events A, B with $P(B) > 0$, the **conditional probability of A given B** is

$$P(A | B) = \frac{P(A \cap B)}{P(B)}.$$

The formula encodes exactly the example: restrict to outcomes in B (the denominator $P(B)$), then ask what fraction also lie in A (the numerator $P(A \cap B)$). In the die case, $P(A \cap B) = P(\{6\}) = 1/6$ and $P(B) = 1/2$, giving $1/3$ as expected.

Common pitfall $P(A | B)$ and $P(B | A)$ are **different** quantities and confusing them is the most consequential error in all of probability — the “prosecutor’s fallacy”. The probability of a positive test **given** disease is not the probability of disease **given** a positive test. Bayes’ theorem below is precisely the rule connecting the two; it is needed exactly because they are not equal.

3.2 The multiplication rule and total probability

Rearranging the definition gives a way to compute the probability of a **sequence** of events:

$$P(A \cap B) = P(B) P(A | B) = P(A) P(B | A).$$

This **multiplication rule** is how we walk down a probability tree, multiplying branch probabilities.

Worked example — two draws without replacement. An urn holds 3 red and 2 green balls. Draw two without replacement. The probability both are red is

$$P(\text{red}_1 \cap \text{red}_2) = P(\text{red}_1) P(\text{red}_2 | \text{red}_1) = \frac{3}{5} \times \frac{2}{4} = \frac{3}{10}.$$

The second factor is conditional: removing a red ball changes the urn.

Often an event can be reached through several disjoint routes. Summing over the routes gives the **law of total probability**.

Theorem 12 (Law of total probability) Let $B_1, \dots, B_n$ be a **partition** of Ω — disjoint events with $B_1 \cup \dots \cup B_n = \Omega$ and each $P(B_i) > 0$. Then for any event A ,

$$P(A) = \sum_{i=1}^n P(A | B_i) P(B_i).$$

Proof. The events $A \cap B_i$ are disjoint (since the B_i are) and their union is $A \cap \Omega = A$. By additivity $P(A) = \sum_i P(A \cap B_i)$, and each $P(A \cap B_i) = P(A | B_i)P(B_i)$ by the multiplication rule. $\square$

In words: to find the overall chance of A , average its conditional chance over each scenario B_i , weighting by how likely that scenario is. The partition is usually the “hidden state” of the problem — which machine made the part, whether the patient is ill, which urn was chosen.

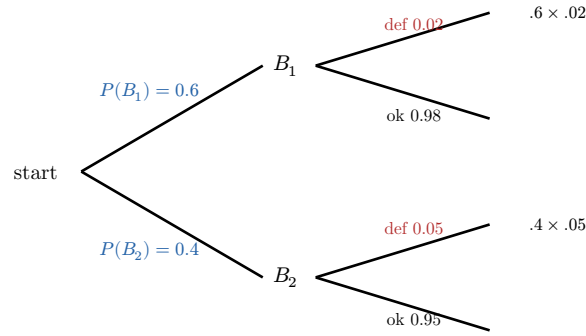


Figure 4: A two-stage experiment. Machine B_1 or B_2 acts first (a partition of Ω), then a defect occurs or not. The probability of a defect is the sum over the two red routes: $0.6(0.02) + 0.4(0.05) = 0.032$ — the law of total probability.

3.3 Bayes' theorem

We now **reverse** the conditioning. We know $P(A | B_i)$ — the chance of the evidence under each scenario — and want $P(B_i | A)$ — which scenario is true given the evidence. The definition of conditional probability, applied twice, supplies the bridge.

Theorem 13 (Bayes' theorem) For a partition $B_1, \dots, B_n$ of Ω and an event A with $P(A) > 0$,

$$P(B_i | A) = \frac{P(A | B_i) P(B_i)}{\sum_{j=1}^n P(A | B_j) P(B_j)}.$$

The numerator is the multiplication rule; the denominator is $P(A)$ by the law of total probability.

The vocabulary we will keep for the rest of the book: $P(B_i)$ is the **prior** (belief before evidence), $P(A | B_i)$ is the **likelihood** (how well scenario i explains the evidence), and $P(B_i | A)$ is the **posterior** (belief after evidence). Bayes' theorem is the rule turning prior into posterior.

Worked example — the surprising medical test. A disease affects 1% of a population. A test is 99% sensitive (it flags 99% of the ill) and 95% specific (it correctly clears 95% of the healthy, so it **false-positives** 5% of the time). You test positive. What is the probability you are actually ill?

Let D = “ill”, D^c = “healthy”, $+$ = “positive”. The given numbers are the prior $P(D) = 0.01$ and the likelihoods $P(+ | D) = 0.99$, $P(+ | D^c) = 0.05$. First the denominator — the overall chance of a positive test:

$$P(+) = P(+|D)P(D) + P(+|D^c)P(D^c) = 0.99(0.01) + 0.05(0.99) = 0.0099 + 0.0495 = 0.0594.$$

Then Bayes:

$$P(D | +) = \frac{0.99 \times 0.01}{0.0594} = \frac{0.0099}{0.0594} \approx 0.167.$$

Only about 17% — despite the “99% accurate” test. The reason is the prior: the ill are so rare that the **false positives from the huge healthy majority** (0.0495) swamp the **true positives from the tiny ill minority** (0.0099). The prior cannot be ignored.

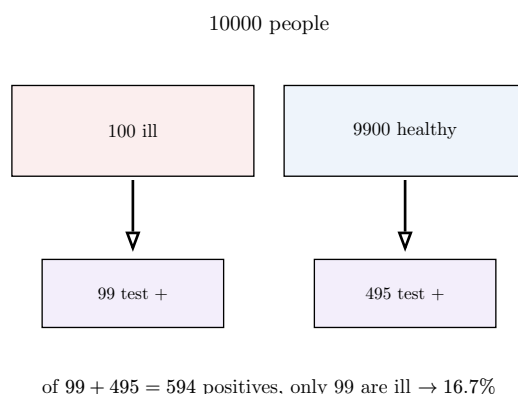


Figure 5: Bayes in natural frequencies. Among 10000 people, 99 true positives arise from the 100 ill, but 495 false positives arise from the 9900 healthy. A positive result is far more often a false alarm — exactly $99/594 \approx 16.7\%$.

The frequency picture is the same computation as Bayes’ theorem, and it is the most reliable way to **explain** a posterior to a non-specialist — and to sanity-check your own algebra.

Exercises

8. Two machines make 60% and 40% of output, with defect rates 2% and 5%. A part is defective; what is the probability machine 2 made it?
9. A standard “Monty Hall” setup: 3 doors, one prize. You pick door 1; the host opens door 3 (no prize) and offers a switch. Use conditional probability to decide whether to switch.
10. Repeat the medical-test example with prevalence $P(D) = 0.10$. How does the posterior change, and why?

4 Independence

The law of total probability handles events that **influence** each other. The opposite case — events that carry no information about each other — is equally fundamental and underlies every “repeated trials” model in the course.

4.1 Independence of two events

Worked example — two coins. Flip two fair coins. Does knowing the first is heads change the chance the second is heads? No — physically separate flips. So $P(\text{second H} \mid \text{first H}) = 1/2 = P(\text{second H})$. Plugging into the multiplication rule, $P(\text{both H}) = 1/2 \times 1/2 = 1/4$, which the four equally likely outcomes HH, HT, TH, TT confirm.

Definition 14 Events A and B are **independent** if

$$P(A \cap B) = P(A)P(B).$$

Equivalently, when $P(B) > 0$, if $P(A \mid B) = P(A)$ — knowing B does not change the probability of A .

We take the **product form** $P(A \cap B) = P(A)P(B)$ as the definition because it is symmetric and remains meaningful even when $P(B) = 0$.

Common pitfall Do not confuse **independent** with **mutually exclusive**. Mutually exclusive events ($A \cap B = \emptyset$) are about as **dependent** as can be: if one occurs the other cannot, so learning A is hugely informative about B . Indeed for disjoint events $P(A \cap B) = 0$, which equals $P(A)P(B)$ only if one of them is impossible. “Disjoint” and “independent” almost never hold together.

4.2 Mutual independence and a famous counterexample

For three or more events, **pairwise** independence (every pair independent) is weaker than **mutual** independence, which demands the product rule for **every sub-collection**.

Definition 15 Events $A_1, \dots, A_n$ are **mutually independent** if for **every** subset $I \subseteq \{1, \dots, n\}$,

$$P\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} P(A_i).$$

In particular all pairs, all triples, etc. must factor.

Worked example — pairwise but not mutually independent (Bernstein). Flip two fair coins. Let A = “first is heads”, B = “second is heads”, C = “the two coins differ”. Each has probability $1/2$. Check the pairs: $P(A \cap B) = P(\text{HH}) = 1/4 = P(A)P(B)$; $P(A \cap C) = P(\text{HT}) = 1/4 = P(A)P(C)$; and likewise B, C . **All three pairs are independent.** But the **triple** fails: if A and B both occur the coins are equal, so C cannot — $P(A \cap B \cap C) = 0$, whereas $P(A)P(B)P(C) = 1/8 \neq 0$. The three events are pairwise but not mutually independent.

The lesson: knowing every **pair** is unrelated does not guarantee the **whole collection** is. For repeated independent trials — the backbone of the binomial and Poisson models — we always require the stronger, mutual, independence.

Random variables X and Y are **independent** when the events they generate are: $P(X = x, Y = y) = P(X = x)P(Y = y)$ for all values x, y . This is the property we need for the variance-of-a-sum formula in the next chapter.

Exercises

11. Show that if A and B are independent, so are A and B^c .
12. Roll two dice. Are “first die is even” and “sum is 7” independent? Compute and decide.
13. Give three events that are mutually independent, and verify all four product conditions.

5 Discrete random variables

A **random variable** attaches a number to each outcome — the sum of two dice, the number of defects in a box, the count of heads in ten flips. It lets us replace clumsy talk of events with the powerful machinery of **expectation** and **variance**.

5.1 Probability mass function and CDF

Worked example — sum of two dice. Let X be the sum of two fair dice. X takes values 2, ..., 12, but not equally often: $X = 7$ arises from 6 of the 36 ordered pairs, while $X = 2$ arises from only 1. The table of $P(X = x)$ is the variable's **fingerprint**.

Definition 16 A **discrete random variable** X takes values in a finite or countable set. Its **probability mass function** (PMF) is $p_X(x) = P(X = x)$. A PMF satisfies $p_X(x) \geq 0$ and $\sum_x p_X(x) = 1$. Its **cumulative distribution function** (CDF) is $F_X(x) = P(X \leq x) = \sum_{t \leq x} p_X(t)$ — a non-decreasing step function rising from 0 to 1.

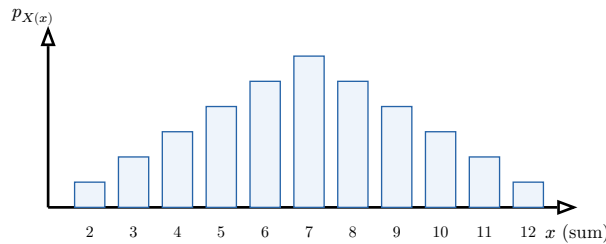


Figure 6: The PMF of the sum of two dice: a triangular shape peaking at 7. The bars are the 11 probabilities $p_{X(2)}, \dots, p_{X(12)}$; they sum to 1. The outcome 7 is six times as likely as 2 because more ordered pairs produce it.

5.2 Expectation

Worked example — a fair price for a game. A game pays €1 if a die shows 1–4 and €10 if it shows 5–6. What is a fair entry fee — the **long-run average** payout? Weight each prize by its probability: $1 \times 4/6 + 10 \times 2/6 = 4/6 + 20/6 = 4$. Over many plays you would pay out €4 per game on average.

Definition 17 The **expectation** (or mean) of a discrete random variable X is the probability-weighted average of its values,

$$\mathbb{E}[X] = \sum_x x p_X(x),$$

provided the sum converges absolutely. We also write $\mu = \mathbb{E}[X]$.

Expectation is **not** the most likely value, nor a value X ever takes (a die's mean is 3.5). It is the **balance point** of the PMF — the centre of mass. A key computational tool: the expectation of a **function** of X needs no new distribution,

$$\mathbb{E}[g(X)] = \sum_x g(x) p_X(x),$$

known as the “law of the unconscious statistician”. We use it constantly with $g(x) = x^2$ to get variance.

5.3 Variance and the König–Huygens formula

Two games can share a mean yet differ wildly in **risk**. A sure €4 and a coin-flip between €0 and €8 both average €4; the second is far more variable. Variance measures that spread.

Definition 18 The **variance** of X is the expected squared deviation from the mean,

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \sum_x (x - \mu)^2 p_X(x) \geq 0.$$

Its square root $\sigma = \sqrt{\text{Var}(X)}$ is the **standard deviation**, in the same units as X .

Expanding the square gives a formula that is almost always easier to compute with.

Proposition 19 (König–Huygens)

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Proof. Expand using linearity (next section) and $\mathbb{E}[\mu] = \mu$, $\mathbb{E}[\mu X] = \mu \mathbb{E}[X] = \mu^2$:

$$\mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] = \mathbb{E}[X^2] - 2\mu \mathbb{E}[X] + \mu^2 = \mathbb{E}[X^2] - \mu^2.$$

□

Worked example — variance of one die. For X uniform on $\{1, \dots, 6\}$, $\mathbb{E}[X] = 3.5$ and $\mathbb{E}[X^2] = (1 + 4 + 9 + 16 + 25 + 36)/6 = 91/6 \approx 15.17$. So $\text{Var}(X) = 91/6 - (7/2)^2 = 91/6 - 49/4 = 35/12 \approx 2.92$, and $\sigma \approx 1.71$.

Common pitfall Variance is in **squared** units (euros², say), so it is not directly comparable to the data; report the **standard deviation** for interpretation. And never forget the $-\mu^2$ in König–Huygens: $\mathbb{E}[X^2] \neq (\mathbb{E}[X])^2$ in general — equality would force variance to zero.

5.4 Linearity of expectation and variance of sums

The single most useful fact about expectation is that it respects sums and scaling **always** — no independence required.

Theorem 20 (Linearity of expectation) For any random variables X, Y and constants a, b ,

$$\mathbb{E}[aX + bY] = a \mathbb{E}[X] + b \mathbb{E}[Y].$$

This holds **whether or not X and Y are independent**.

Worked example — expected heads in 10 flips, the easy way. Let X be the number of heads in 10 fair flips. Writing $X = X_1 + \dots + X_{10}$ where X_i is 1 if flip i is heads and 0 otherwise, each $\mathbb{E}[X_i] = 1/2$, so $\mathbb{E}[X] = 10 \times 1/2 = 5$ — no binomial sum needed. This **indicator trick** is the workhorse of the next chapter.

Variance is **not** linear. It scales by the square and only adds across **independent** variables.

Theorem 21 (Variance of a sum) For any X and constant a , $\text{Var}(aX + b) = a^2 \text{Var}(X)$. If X and Y are **independent**, then

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

Common pitfall The additivity $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ **requires independence**; in general there is a covariance correction term. Adding a constant shifts the mean but leaves variance unchanged ($\text{Var}(X + b) = \text{Var}(X)$) — spread does not care where the centre is. And multiplying by a multiplies variance by a^2 , not a .

Exercises

14. A variable takes values $-1, 0, 2$ with probabilities $0.2, 0.5, 0.3$. Find $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, and $\text{Var}(X)$.
15. Using indicators, find the expected number of fixed points when n people's hats are returned at random (the expected number who get their own hat back).
16. X, Y independent with variances 4 and 9. Find $\text{Var}(2X - Y + 5)$.

6 Classic discrete distributions

A handful of distributions describe most discrete random phenomena. The point of this chapter is **not** to memorise four formulas but to learn the **modelling assumption** behind each, so you can recognise — and if needed re-derive — the right one. Each distribution is the inevitable answer to a specific question.

6.1 Bernoulli: a single yes/no trial

Definition 22 A **Bernoulli** variable $X \sim \text{Ber}(p)$ models one trial with two outcomes: $X = 1$ (“success”) with probability p and $X = 0$ (“failure”) with probability $1 - p$. Then

$$\mathbb{E}[X] = p, \quad \text{Var}(X) = p(1 - p).$$

The mean is immediate; for the variance, $\mathbb{E}[X^2] = p$ (since $X^2 = X$ for a 0/1 variable), so $\text{Var}(X) = p - p^2 = p(1 - p)$. Note the spread is largest at $p = 1/2$ (maximum uncertainty) and vanishes at $p = 0$ or 1 (certainty). The Bernoulli is the atom from which the next two distributions are built.

6.2 Binomial: counting successes in n independent trials

Worked example — defects in a box. A machine produces a defective part with probability $p = 0.02$, independently each time. In a box of $n = 50$, let X count the defects. The chance of **exactly** k defects must account for **which** k of the 50 parts are bad: any particular pattern with k defects has probability $p^k(1 - p)^{n-k}$, and there are $\binom{n}{k}$ such patterns.

Definition 23 If X is the number of successes in n **independent** Bernoulli(p) trials, then $X \sim \text{Bin}(n, p)$ with PMF

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n.$$

These probabilities sum to 1 by the binomial theorem: $\sum_{k=0}^n \binom{n}{k} p^k (1 - p)^{n-k} = (p + (1 - p))^n = 1$ — the identity from Chapter 1 doing exactly the bookkeeping it was built for. Because $X = X_1 + \dots + X_n$ is a sum of independent Bernoulli(p) variables, linearity and variance-additivity give the moments **instantly**:

$$\mathbb{E}[X] = np, \quad \text{Var}(X) = np(1 - p).$$

For the box, $\mathbb{E}[X] = 50 \times 0.02 = 1$ expected defect, and $P(X = 0) = (0.98)^{50} \approx 0.364$, so $P(X \geq 1) \approx 0.636$ — a defect is more likely than not.

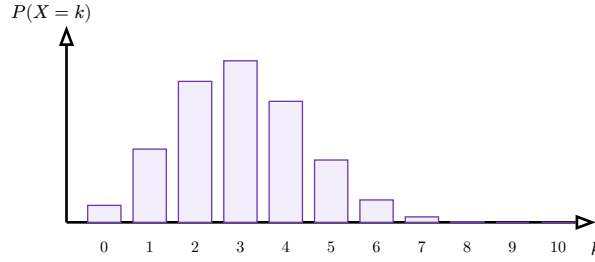


Figure 7: The binomial PMF for $\text{Bin}(10, 0.3)$. The distribution peaks near the mean $np = 3$ and is right-skewed because $p < 0.5$. As n grows it becomes bell-shaped — a preview of the Central Limit Theorem.

6.3 Geometric: waiting for the first success

Worked example — first six. You roll a die repeatedly until the first 6. The probability the **first** 6 appears on roll k requires $k - 1$ non-sixes (each probability $5/6$) followed by a six: $(5/6)^{k-1}(1/6)$.

Definition 24 If X is the number of independent Bernoulli(p) trials **up to and including** the first success, then $X \sim \text{Geom}(p)$ with PMF

$$P(X = k) = (1 - p)^{k-1}p, \quad k = 1, 2, 3, \dots$$

These sum to 1 by the **geometric series** of MAT11-1: $\sum_{k=1}^{\infty} (1 - p)^{k-1}p = p/(1 - (1 - p)) = 1$. Differentiating that series (or a short algebraic argument) gives the moments

$$\mathbb{E}[X] = \frac{1}{p}, \quad \text{Var}(X) = \frac{1 - p}{p^2}.$$

The mean is wonderfully intuitive: if a success happens one time in p , you wait on average $1/p$ trials. For the die, $\mathbb{E}[X] = 6$.

The geometric distribution is **memoryless**: $P(X > m + n \mid X > m) = P(X > n)$ — having waited m failures, the future is statistically fresh. The die has no memory of its past misses, and the geometric is the **only** discrete distribution with this property.

6.4 Poisson: counts of rare events, as a limit of the binomial

Worked example — typos on a page. A proofreader finds on average $\lambda = 2$ typos per page. Typos are many possible (n large — every word could be wrong) but each individually rare (p small), with $np = \lambda$ fixed. What is the distribution of the count? Take the binomial $\text{Bin}(n, \lambda/n)$ and let $n \rightarrow \infty$.

Proposition 25 (Poisson limit of the binomial) If $n \rightarrow \infty$ and $p \rightarrow 0$ with $np \rightarrow \lambda$ fixed, then

$$\binom{n}{k} p^k (1-p)^{n-k} \rightarrow e^{-\lambda} \frac{\lambda^k}{k!}.$$

Proof. Write $p = \lambda/n$. Then

$$\binom{n}{k} p^k (1-p)^{n-k} = \frac{n(n-1)\cdots(n-k+1)}{k!} \cdot \frac{\lambda^k}{n^k} \cdot \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-k}.$$

As $n \rightarrow \infty$: the falling product over $n^k \rightarrow 1$; the factor $(1 - \lambda/n)^n \rightarrow e^{-\lambda}$ (the limit defining the exponential from MAT11-1); and $(1 - \lambda/n)^{-k} \rightarrow 1$. What survives is $\lambda^k/k! \cdot e^{-\lambda}$. $\square$

Definition 26 A **Poisson** variable $X \sim \text{Poi}(\lambda)$ models the count of independent rare events with mean rate $\lambda > 0$:

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots,$$

with the striking property that mean and variance coincide:

$$\mathbb{E}[X] = \lambda, \quad \text{Var}(X) = \lambda.$$

The probabilities sum to 1 because $\sum_{k=0}^{\infty} \lambda^k/k! = e^{\lambda}$ (the exponential series), cancelling the $e^{-\lambda}$ prefactor. The Poisson governs calls to a switchboard, decays of a radioactive sample, defects per batch, arrivals in a queue — anything that counts rare independent events over a fixed exposure.

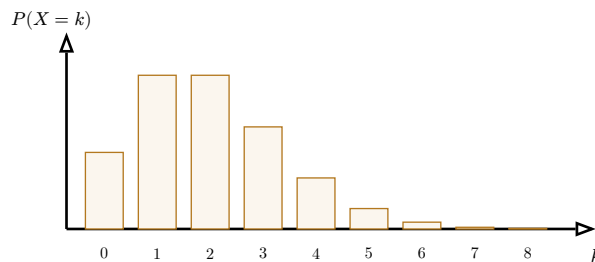


Figure 8: The Poisson PMF for $\lambda = 2$ (mean typos per page). Mass concentrates near λ , with a long right tail. Here mean and variance are both 2 — a signature used to **test** whether data are really Poisson.

Common pitfall Choosing a distribution is a **modelling decision**, not pattern-matching on numbers. Ask: is this **one** trial (Bernoulli), a **fixed number** of trials counting successes (binomial), a **wait** for the first success (geometric), or a **count of rare events** over a fixed exposure (Poisson)? The wrong assumption — e.g. treating dependent draws as independent, or using Poisson when the rate is not constant — invalidates the formula however carefully you compute it.

Reading a problem into a distribution.

- one yes/no trial $\rightarrow$ **Bernoulli**(p) — mean p , variance $p(1-p)$;
- successes in n fixed independent trials $\rightarrow$ **Binomial**(n, p) — mean np , variance $np(1-p)$;
- trials until first success $\rightarrow$ **Geometric**(p) — mean $1/p$, variance $(1-p)/p^2$;
- count of rare independent events, rate $\lambda \rightarrow$ **Poisson**(λ) — mean and variance both λ .

6.5 Building a model from a word problem

The central skill of this book — **reasoning over recognition** — deserves an explicit recipe. Given a word problem, before any arithmetic:

1. **Name the experiment and the sample space Ω .** What is one outcome?
2. **Define the random variable X** — exactly what does it count or measure?
3. **Check the modelling assumptions.** Are trials independent? Is the success probability constant? Is the number of trials fixed or are we waiting?
4. **Choose the distribution** the assumptions force, and only then write the PMF.
5. **Compute, then sanity-check** against intuition (is the mean plausible? is a probability between 0 and 1?).

Worked example — a full model. “A call centre receives on average 3 calls per minute. What is the probability of at least 2 calls in a given minute?” The events are calls; we **count** them over a fixed one-minute exposure; calls are many-possible and individually rare, arriving independently at a constant rate — the Poisson assumptions. So $X \sim \text{Poi}(3)$. Then $P(X \geq 2) = 1 - P(X = 0) - P(X = 1) = 1 - e^{-3} - 3e^{-3} = 1 - 4e^{-3} \approx 0.801$. The model choice did the real work; the formula was the easy part.

Exercises

17. A fair coin is flipped 8 times. Find $P(\text{exactly 3 heads})$ and the expected number of heads.
18. Cards are drawn **with** replacement from a deck until the first ace. What is the expected number of draws? Which distribution, and why?
19. A website gets 240 visits per hour on average. Model the number of visits in a given minute and find the probability of more than 5 in one minute.
20. Explain why “the number of defective items in a shipment of 100, drawn **without** replacement from a finite warehouse” is only **approximately** binomial.

7 Introduction to estimation

Until now the distribution’s parameter — p, λ — was **given**. In practice it is **unknown** and must be **estimated from data**. This chapter sets up the language; the next computes the estimates.

7.1 Estimators, bias, and mean squared error

Worked example — estimating a coin's bias. You flip an unknown coin 100 times and see 58 heads. A natural guess for p is $\hat{p} = 58/100 = 0.58$ — the sample proportion. But “natural” is not “correct”: $\hat{p}$ is itself random (another 100 flips would give a different number), so we must study its **behaviour as a random variable**.

Definition 27 Given data $X_1, \dots, X_n$ from a distribution with unknown parameter θ , an **estimator** $\hat{\theta} = \hat{\theta}(X_1, \dots, X_n)$ is any function of the data used to approximate θ . Being a function of random data, $\hat{\theta}$ is itself a random variable with its own distribution (its **sampling distribution**).

Two questions judge an estimator. **On average, does it hit the target?** — that is **bias**. **How tightly does it cluster?** — that is **variance**. Mean squared error combines both.

Definition 28 The **bias** of $\hat{\theta}$ is $\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$. The estimator is **unbiased** if $\text{Bias}(\hat{\theta}) = 0$, i.e. $\mathbb{E}[\hat{\theta}] = \theta$. The **mean squared error** is

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2].$$

Theorem 29 (bias–variance decomposition)

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2.$$

Proof. Let $m = \mathbb{E}[\hat{\theta}]$, so $\text{Bias} = m - \theta$. Insert $\pm m$:

$$\mathbb{E}[(\hat{\theta} - \theta)^2] = \mathbb{E}[(\hat{\theta} - m + (m - \theta))^2].$$

Expanding, the cross term is $2(m - \theta)\mathbb{E}[\hat{\theta} - m] = 0$ because $\mathbb{E}[\hat{\theta} - m] = 0$. The two square terms are $\text{Var}(\hat{\theta})$ and $(m - \theta)^2 = \text{Bias}^2$. $\square$

This decomposition is one of the most important ideas in statistics: error has two sources, **systematic** (bias) and **random** (variance), and a good estimator balances them. An unbiased estimator has $\text{MSE} = \text{Var}(\hat{\theta})$, so among unbiased estimators we prefer the one with smallest variance.

Worked example — the sample proportion is unbiased. For coin flips, $\hat{p} = (1/n) \sum_{i=1}^n X_i$ with $X_i \sim \text{Ber}(p)$. By linearity $\mathbb{E}[\hat{p}] = (1/n) \sum \mathbb{E}[X_i] = (1/n)(np) = p$, so $\hat{p}$ is **unbiased**. Its variance is $\text{Var}(\hat{p}) = (1/n^2) \sum \text{Var}(X_i) = (1/n^2)(np(1-p)) = p(1-p)/n$, which shrinks like $1/n$ — more data, tighter estimate. Here $\text{MSE} = p(1-p)/n$.

Common pitfall Unbiased is not the same as accurate on one sample. An unbiased estimator can be far off on any single dataset; unbiasedness is a statement about the **average over many hypothetical datasets**. Conversely a slightly biased estimator with much smaller variance can have lower MSE — bias is not automatically bad.

8 Maximum likelihood estimation

Where do estimators **come from**? The sample proportion felt natural, but we want a principled, general recipe. **Maximum likelihood** supplies one: choose the parameter value under which the data we actually observed were **most probable**.

8.1 The likelihood and log-likelihood

Worked example — which coin?. You flip a coin 3 times and see H, H, T. Was $p = 0.5$ or $p = 0.8$? Under $p = 0.5$ this sequence has probability $0.5^2 \times 0.5 = 0.125$; under $p = 0.8$, $0.8^2 \times 0.2 = 0.128$. The data are (just) more probable under $p = 0.8$, so likelihood prefers it. Maximum likelihood pushes this idea to its logical end: find the p that maximises the probability of the observed data.

Definition 30 Given observed data $x_1, \dots, x_n$ from a distribution with PMF $p(x; \theta)$, the **likelihood function** is their joint probability **viewed as a function of θ** ,

$$L(\theta) = \prod_{i=1}^n p(x_i; \theta),$$

(the product assumes independent observations). The **maximum likelihood estimator** (MLE) is the value $\hat{\theta}$ maximising $L(\theta)$.

Products are awkward to differentiate, so we maximise the **log-likelihood** $\ell(\theta) = \ln L(\theta) = \sum_{i=1}^n \ln p(x_i; \theta)$ instead. Because $\ln$ is strictly increasing, it has the **same** maximiser — but turns products into sums, which the calculus of MAT11-1 handles by setting $\ell'(\theta) = 0$.

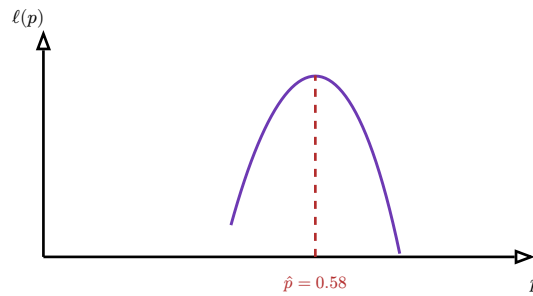


Figure 9: The log-likelihood for 58 heads in 100 flips. It is concave with a single peak; the maximiser $\hat{p} = 0.58$ — the sample proportion — is found by solving $\ell'(p) = 0$.

8.2 MLE for the classic distributions

Worked example — Bernoulli / binomial MLE. Observe $X_1, \dots, X_n \sim \text{Ber}(p)$ with $s = \sum x_i$ successes. The log-likelihood is

$$\ell(p) = \sum_i [x_i \ln p + (1 - x_i) \ln(1 - p)] = s \ln p + (n - s) \ln(1 - p).$$

Differentiate and set to zero:

$$\ell'(p) = \frac{s}{p} - \frac{n - s}{1 - p} = 0 \Rightarrow s(1 - p) = (n - s)p \Rightarrow \hat{p} = \frac{s}{n}.$$

The MLE is the **sample proportion** — vindicating the “natural” guess. The second derivative is negative, confirming a maximum. (A single $\text{Bin}(n, p)$ observation k gives the same $\hat{p} = k/n$.) We showed above this estimator is **unbiased**.

Worked example — Poisson MLE. Observe counts $X_1, \dots, X_n \sim \text{Poi}(\lambda)$. Since $p(k; \lambda) = e^{-\lambda} \lambda^k / k!$,

$$\ell(\lambda) = \sum_i (-\lambda + x_i \ln \lambda - \ln(x_i!)) = -n\lambda + \left(\sum_i x_i \right) \ln \lambda - \text{const}.$$

Then $\ell'(\lambda) = -n + (\sum_i x_i) / \lambda = 0$ gives

$$\hat{\lambda} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x},$$

the **sample mean**. It is unbiased: $\mathbb{E}[\hat{\lambda}] = (1/n) \sum \mathbb{E}[X_i] = (1/n)(n\lambda) = \lambda$. For example, observing 3, 1, 4, 2, 0 typos on five pages gives $\hat{\lambda} = 10/5 = 2$ typos per page.

Common pitfall Two routine slips. First, **always take the log before differentiating** — attacking the raw product $L(\theta)$ is far harder and error-prone. Second, **check it is a maximum, not a minimum**, and that the solution lies in the parameter’s valid range (a probability in $[0, 1]$, a rate $\lambda > 0$). The first-order condition $\ell'(\theta) = 0$ is necessary, not sufficient.

Exercises

21. Derive the Poisson MLE from scratch, including the second-derivative check.
22. You observe 7 successes in 20 independent trials. Give the MLE of p and the estimated variance of $\hat{p}$.
23. For a geometric(p) sample $x_1, \dots, x_n$, show the MLE is $\hat{p} = n / \sum_i x_i = 1/\bar{x}$.

9 Bayesian updating

Maximum likelihood reads a parameter **off the data alone**. Bayesian inference adds a second ingredient: a **prior** belief, updated by data into a **posterior**. We met the mechanism — Bayes’ theorem — for events; now we apply it to **parameters**, and watch beliefs move as evidence accumulates.

9.1 Prior, likelihood, posterior

Worked example — is the factory line faulty? A production line is “normal” (defect rate 0.01) or “faulty” (rate 0.10). Experience says faults are rare: prior $P(\text{faulty}) = 0.05$, $P(\text{normal}) = 0.95$. You inspect one item and it is **defective**. Update:

$$P(\text{faulty} \mid \text{def}) = \frac{P(\text{def} \mid \text{faulty})P(\text{faulty})}{P(\text{def} \mid \text{faulty})P(\text{faulty}) + P(\text{def} \mid \text{normal})P(\text{normal})} = \frac{0.10 \times 0.05}{0.10 \times 0.05 + 0.01 \times 0.95} = \frac{0.005}{0.0145} \approx 0.345.$$

One defect lifts the worry from 5% to 34.5%. A **second** defect would update again, using 0.345 as the new prior — beliefs compound.

Definition 31 To estimate a parameter θ the Bayesian way: specify a **prior** distribution $\pi(\theta)$ (belief before data); compute the **likelihood** $L(\theta) = P(\text{data} \mid \theta)$; and form the **posterior** by Bayes’ theorem,

$$\pi(\theta \mid \text{data}) = \frac{L(\theta) \pi(\theta)}{P(\text{data})} \propto L(\theta) \pi(\theta),$$

where $P(\text{data}) = \sum_{\theta} L(\theta) \pi(\theta)$ (or an integral for continuous θ) is the normalising constant. In words: **posterior** $\propto$ **likelihood** $\times$ **prior**.

9.2 How data update beliefs — a worked continuous example

When the parameter is a **continuous** probability $p \in [0, 1]$, a convenient prior is the **Beta** family, $\pi(p) \propto p^{a-1}(1-p)^{b-1}$. Its shape is governed by a, b : Beta(1, 1) is flat (total ignorance), while

larger a, b concentrate belief. The Beta is **conjugate** to binomial data — the posterior is again a Beta — which makes updating a matter of **counting**.

Proposition 32 (Beta–Binomial updating) With prior $p \sim \text{Beta}(a, b)$ and data of s successes in n trials, the posterior is

$$p \mid \text{data} \sim \text{Beta}(a + s, b + n - s).$$

Proof. The likelihood is $L(p) \propto p^s(1-p)^{n-s}$ and the prior $\pi(p) \propto p^{a-1}(1-p)^{b-1}$. Their product is $p^{a+s-1}(1-p)^{b+n-s-1}$, the kernel of a $\text{Beta}(a + s, b + n - s)$. Normalisation is automatic for a known family. $\square$

So observing a success simply **increments a** , a failure increments b — the prior acts like a stock of “imaginary” prior trials ($a - 1$ successes, $b - 1$ failures) that the data add to. The posterior mean is

$$\mathbb{E}[p \mid \text{data}] = \frac{a + s}{a + b + n},$$

a **blend** of the prior mean $a/(a + b)$ and the data proportion s/n .

Worked example — watching the posterior shift. A new supplier’s defect probability p is unknown. Start with a weak, slightly-optimistic prior $\text{Beta}(2, 8)$ (prior mean 0.2). Inspect batches:

- After 1 defect in 10: posterior $\text{Beta}(3, 17)$, mean $3/20 = 0.15$.
- After 5 defects in 50 total: posterior $\text{Beta}(7, 53)$, mean $7/60 \approx 0.117$.
- After 50 defects in 500: posterior $\text{Beta}(52, 458)$, mean ≈ 0.102 .

The estimate marches from the prior’s 0.20 toward the data proportion 0.10, and the posterior narrows (variance falls) as n grows. The **prior matters when data are scarce** and is **overwhelmed once data are plentiful** — exactly the behaviour we want from honest inference.

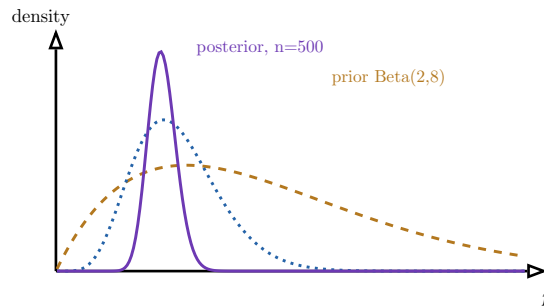


Figure 10: Prior-to-posterior shift for the supplier’s defect rate. The broad prior (dashed) concentrates into an ever-narrower posterior as data accumulate (dotted then solid), homing in on the true rate ≈ 0.10 and shrinking in width — more evidence, less uncertainty.

Common pitfall Bayesian and likelihood answers **agree when data dominate** and **diverge when data are scarce** — that is a feature, not a bug. With little data the prior carries real weight, so a **badly chosen prior** biases the conclusion; with abundant data the likelihood overwhelms any reasonable prior and the two approaches converge. State your prior explicitly so others can judge it.

Three ways to pin down a parameter, in one view.

- **MLE**: maximise $L(\theta)$ — uses data only, no prior. Answer: a single number $\hat{\theta}$.
- **Bayesian**: combine prior $\pi(\theta)$ with likelihood to get a full posterior distribution $\pi(\theta \mid \text{data})$ — answer is a **distribution**, quantifying remaining uncertainty.
- They **coincide** (the posterior peaks at the MLE) under a flat prior and abundant data.

Exercises

- 24.** Redo the faulty-line example after a **second** consecutive defect. Does the posterior cross 50%?
- 25.** With prior $\text{Beta}(1, 1)$ (uniform) and data 3 successes in 10, give the posterior and its mean. Compare with the MLE $\hat{p} = 0.3$.
- 26.** A test is 90% sensitive and 90% specific for a condition with prevalence 2%. Compute the posterior probability of the condition after one positive result, then after a second independent positive result.
- 27.** Explain, using the posterior-mean blend $(a + s)/(a + b + n)$, why the prior's influence vanishes as $n \rightarrow \infty$.